

PROJECT REPORT

AI Meeting Action-Item Extractor

Intelligent Meeting Transcript Analysis, Accountability Tracking &

Automated Action-Item Extraction

Submitted by

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1. Project Title

AI Meeting Action-Item Extractor

Subtitle: Intelligent Meeting Transcript Analysis, Accountability Tracking and Automated Action-Item Extraction

2. Abstract

Meetings generate valuable information such as assigned tasks, decisions, responsibilities, and deadlines. However, manually reviewing meeting discussions and converting them into organized action items requires considerable time and may lead to overlooked or incomplete tasks.

The **AI Meeting Action-Item Extractor** is an intelligent system that uses **Artificial Intelligence (AI) and Natural Language Processing (NLP)** to analyze meeting transcripts and automatically identify actionable information. It extracts key details including the task description, assigned person, deadline, priority, current status, and confidence level.

The system also supports **tasktracking and accountability** by monitoring action items across multiple meetings. It can identify overdue tasks, repeated or postponed responsibilities, and generate follow-up reminders for pending activities. An interactive dashboard provides a clear view of team workload, task progress, deadlines, and meeting outcomes.

Validation mechanisms are incorporated to detect missing task owners, unavailable deadlines, duplicate action items, and low-confidence extractions. By converting unstructured meeting conversations into structured and actionable information, the proposed system helps teams improve task visibility, accountability, and post-meeting productivity.

3. Problem Statement

Meetings often generate a large amount of unstructured information, making it difficult for participants to identify and organize assigned tasks, responsible team members, deadlines, and pending decisions after the discussion. Manually reviewing meeting transcripts to extract these details requires significant time and can lead to missed or incorrectly recorded action items.

In addition, conventional meeting-management systems generally focus on capturing information from individual meetings without maintaining continuous tracking of tasks across multiple meetings. As a result, completed, overdue, or repeatedly postponed tasks may not be identified effectively, creating gaps in project accountability and follow-up.

Therefore, there is a need for an **AI-powered system** that can understand meeting conversations, automatically extract and structure actionable information, validate the extracted details, and continuously monitor task progress across meetings. Such a system can improve task visibility, reduce manual effort, strengthen accountability, and support effective post-meeting follow-up.

4. Objectives

The main objectives of this project are:

1. To process and understand meeting transcripts using Natural Language Processing techniques.
2. To automatically detect and extract actionable tasks from meeting discussions.
3. To identify the team member responsible for completing each task.
4. To recognize deadlines and due dates mentioned during the meeting.

5. To determine the current status of each identified action item.
6. To calculate a confidence score for the extracted task information.
7. To validate extracted data and detect missing, incomplete, or inconsistent fields.
8. To identify similar or duplicate action items to avoid repeated task entries.
9. To classify action items according to their priority using urgency-related terms and context.
10. To maintain action-item history across multiple meetings and identify overdue or repeatedly delayed tasks.
11. To generate follow-up reminder drafts for pending or overdue action items.
12. To provide an interactive dashboard for viewing tasks, deadlines, priorities, and team workload.
13. To allow users to review, modify, and confirm AI-generated results when necessary.

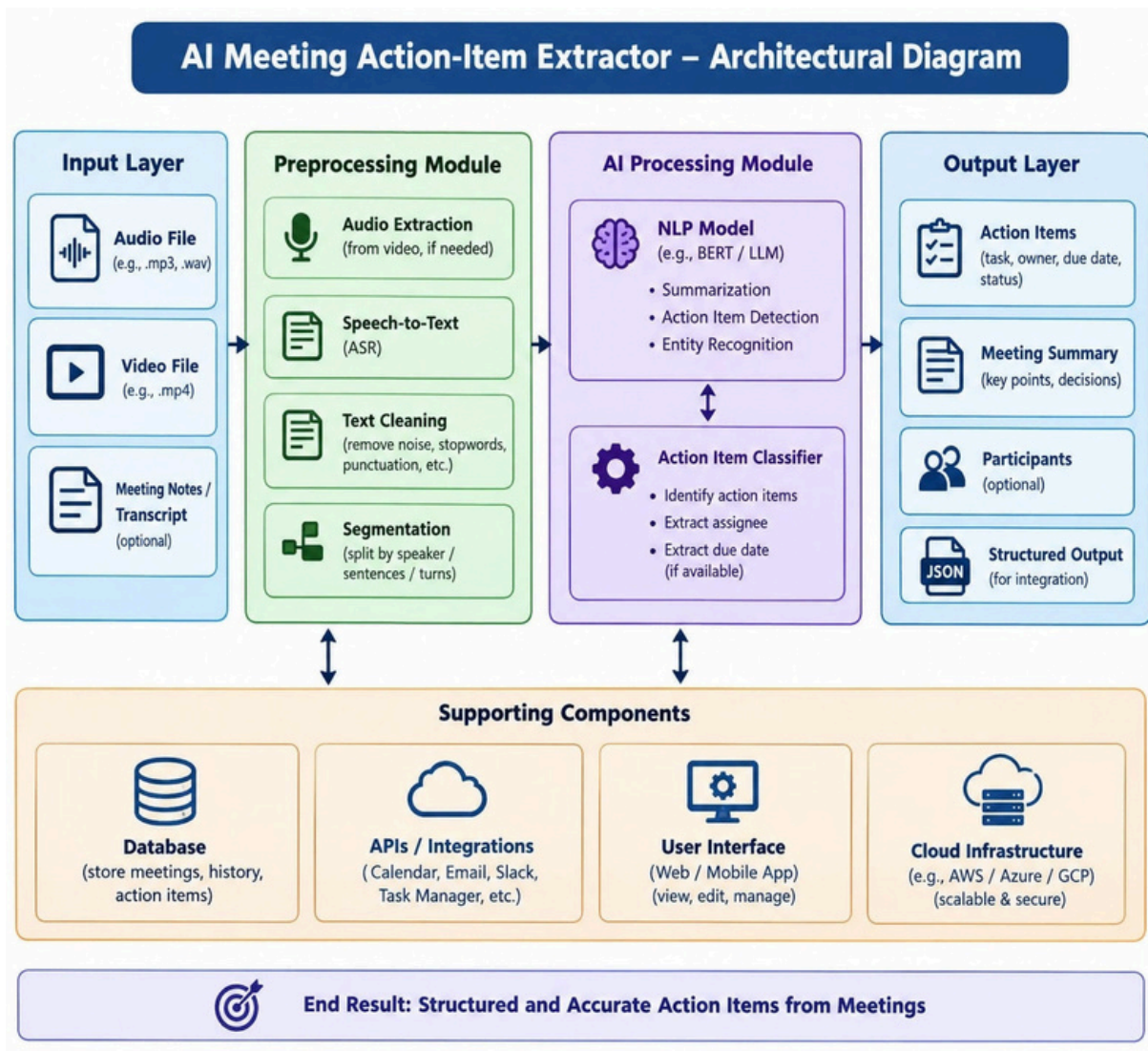
5. Proposed System

The proposed system accepts a meeting transcript as input and processes the conversation using NLP/AI techniques. It performs the following operations:

Meeting Transcript → Text Preprocessing → Speaker/Sentence Segmentation → AI Information Extraction → Action-Item Identification → Task/Owner/Deadline/Status/Priority Extraction → Validation → Confidence Score → Cross-Meeting Tracking → Dashboard & Follow-Up Generation

The final output presents the extracted action items in a structured, filterable table, along with a workload dashboard and auto-drafted reminder messages for pending items.

6. System Architecture



7. Modules Used

1. User Authentication Module

- Provides secure login and user access.
- Manages user accounts and sessions.

2. Meeting Transcript Upload Module

- Allows users to upload meeting transcripts.

- Supports formats such as TXT, DOCX, and PDF.

3. Text Preprocessing Module

- Removes unnecessary symbols, formatting, and unwanted text.
- Divides the transcript into sentences and speaker segments.

4. NLP Processing Module

- Uses Natural Language Processing to understand meeting conversations.
- Identifies important sentences and task-related information.

5. Action-Item Extraction Module

- Detects tasks and decisions from the transcript.
- Extracts the actual action that needs to be performed.

6. Task Information Extraction Module

- Identifies:
 - Task
 - Responsible person
 - Deadline
 - Status
 - Priority

7. Confidence & Validation Module

- Calculates confidence scores for AI-generated results.
- Detects missing or inconsistent information.

- Sends uncertain results for human verification.

8. Duplicate & Task Tracking Module

- Detects duplicate or similar action items.
- Tracks tasks across multiple meetings.
- Identifies overdue and repeatedly postponed tasks.

9. Reminder Generation Module

- Generates follow-up reminder drafts.
- Helps users communicate pending or overdue tasks.

10. Dashboard & Analytics Module

- Displays action items in a structured table.
- Shows workload, task status, priorities, deadlines, and overdue tasks.
- Provides visual analytics and charts.

11. Human Verification Module

- Allows users to review and correct AI-extracted information.
- Users can approve, edit, or reject uncertain action items.

12. Database & History Module

- Stores meeting transcripts, extracted tasks, users, and task history.
- Maintains records across multiple meetings.

8. Technologies Used

Technology	Purpose
Python	Main programming language for AI/NLP development
Natural Language Processing (NLP)	Understands meeting conversations and identifies actionable information
Machine Learning	Classifies and extracts action items from transcripts
NLTK / spaCy	Text preprocessing, sentence segmentation, and linguistic analysis
Transformers / BERT	Advanced understanding of meeting context and task extraction
Pandas	Organizing and processing extracted task data
Scikit-learn	Classification, similarity detection, and confidence analysis

SQLite / MySQL	Storing users, meetings, action items, and task history
Streamlit	Building the interactive web dashboard
HTML & CSS	Designing and styling the user interface
Plotly / Matplotlib	Creating workload, status, priority, and deadline visualizations
Git & GitHub	Version control and project management

9. Key Features

Core Features

1. Automatic Action-Item Extraction

- Identifies tasks and important actions automatically from meeting transcripts.

2. Task Details Extraction

- Extracts the **task, responsible person, deadline, status, and priority**.

3. NLP-Based Conversation Analysis

- Understands meeting conversations and identifies task-related statements.

4. Confidence Scoring

- Provides a confidence score for each extracted action item.

5. Information Validation

- Detects missing owners, deadlines, incomplete details, and inconsistent information.

6. Duplicate Task Detection

- Identifies similar or repeated action items to prevent duplicate entries.

7. Priority Classification

- Assigns priority levels based on urgency and context in the conversation.

8. Cross-Meeting Task Tracking

- Tracks action items across multiple meetings and maintains their history.

9. Overdue & Repeated Task Detection

- Highlights tasks that are overdue or repeatedly postponed.

10. Follow-Up Reminder Generation

- Creates ready-to-use reminder drafts for pending and overdue tasks.

11. Interactive Dashboard

- Displays tasks, deadlines, priorities, status, and team workload through tables and charts.

12. Human Verification

- Allows users to review, edit, approve, or correct uncertain AI-generated results.

13. Task History Management

- Maintains records of previous meetings and their associated action items.

10. Advantages

Reduces Manual Work – Automatically extracts tasks from meeting transcripts.

Saves Time – Reduces the time required to review long meeting conversations.

Improves Task Visibility – Clearly displays tasks, owners, deadlines, and status.

Reduces Missed Tasks – Helps identify important action items that may be overlooked.

Improves Accountability – Tracks who is responsible for each task.

Continuous Task Tracking – Monitors action items across multiple meetings.

Detects Overdue Tasks – Highlights pending and repeatedly postponed tasks.

Provides Confidence Scores – Helps users identify information that requires verification.

Easy Follow-Up – Generates reminder drafts for pending tasks.

User-Friendly Dashboard – Provides clear tables and visual analytics for quick understanding.

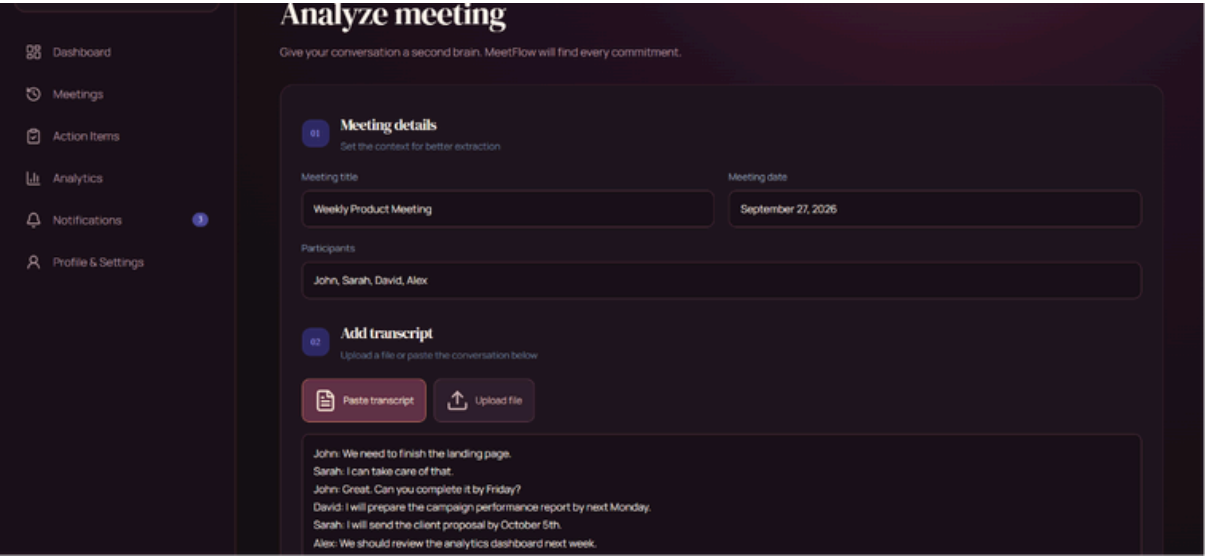
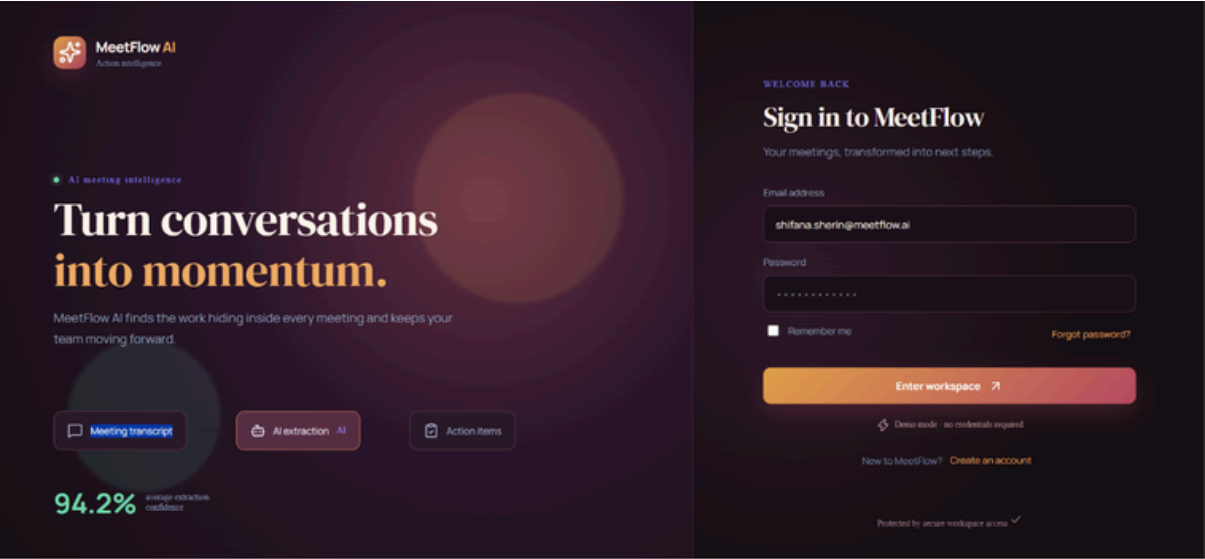
11. Limitations

- **Transcript Quality Dependency** – Poor or incomplete

transcripts can reduce extraction accuracy.

- **Language Limitations** – Performance may vary for different languages, accents, and mixed-language conversations.
 - **Context Understanding** – AI may misunderstand tasks when the conversation is ambiguous.
 - **Incorrect Speaker Identification** – Similar names or unclear speaker information may cause errors in assigning tasks.
 - **Deadline Ambiguity** – Informal phrases such as “next week” or “as soon as possible” may be difficult to interpret accurately.
 - **False Action Items** – The system may occasionally identify general statements as tasks.
 - **Human Verification Required** – Low-confidence or complex results may still require manual review.
 - **Data Privacy Concerns** – Meeting transcripts may contain confidential or sensitive business information.
- Model Dependency** – Extraction quality depends on the selected NLP/ML model and training data.
- Resource Requirements** – Advanced NLP models may require additional computing resources and processing time.

12.Output Screenhot



MeetFlow AI

Action intelligence

Shifana Sherin

Adidas workspace

Dashboard

Meetings

Action Items

Analytics

Notifications

Profile & Settings

AI workspace secured

Workspace / Meetings

All changes saved

Shifana Sherin

MEETING HISTORY

Meetings

Every conversation, one searchable workspace.

Search meetings, participants, transcripts...

Filters

MEETING	DATE	ITEMS	CONFIDENCE	STATUS	ACTIONS
Weekly Product Meeting John, Sarah, David, Alex	September 27, 2026	6 0% complete	85%	Needs review	
Weekly Product Meeting John, Sarah, David, Alex	Sep 26, 2026	3 0% complete	90%	Needs review	
Marketing Strategy Meeting Maya, John, Pops	Sep 24, 2026	2 100% complete	95%	Complete	
Engineering Standup Alex, Nina, Oscar	Sep 22, 2026	1 0% complete	68%	Needs review	

MeetFlow AI

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Shifana Sherin

ACTION INTELLIGENCE

Action items

The commitments extracted from every meeting.

Search tasks, owners, deadlines...

All

TASK	OWNER	DEADLINE	STATUS	CONFIDENCE
Finish the landing page Weekly Product Meeting Human review recommended	John	No deadline	Pending	68%
Take care of that Weekly Product Meeting Human review recommended	Sarah	No deadline	Pending	68%
Great. Can you complete it Weekly Product Meeting	John	Friday?	Pending	95%

EDIT ACTION ITEM

Refine the next step

×

Task

Finish the landing page

Owner

John

Deadline

No deadline

Status

Pending

Source text

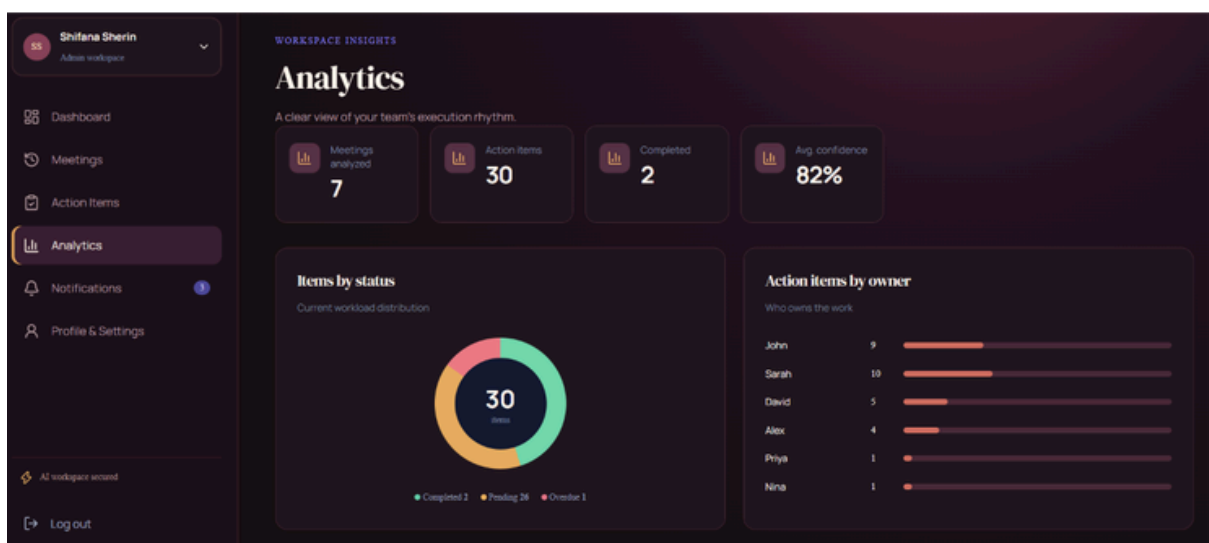
John: We need to finish the landing page.

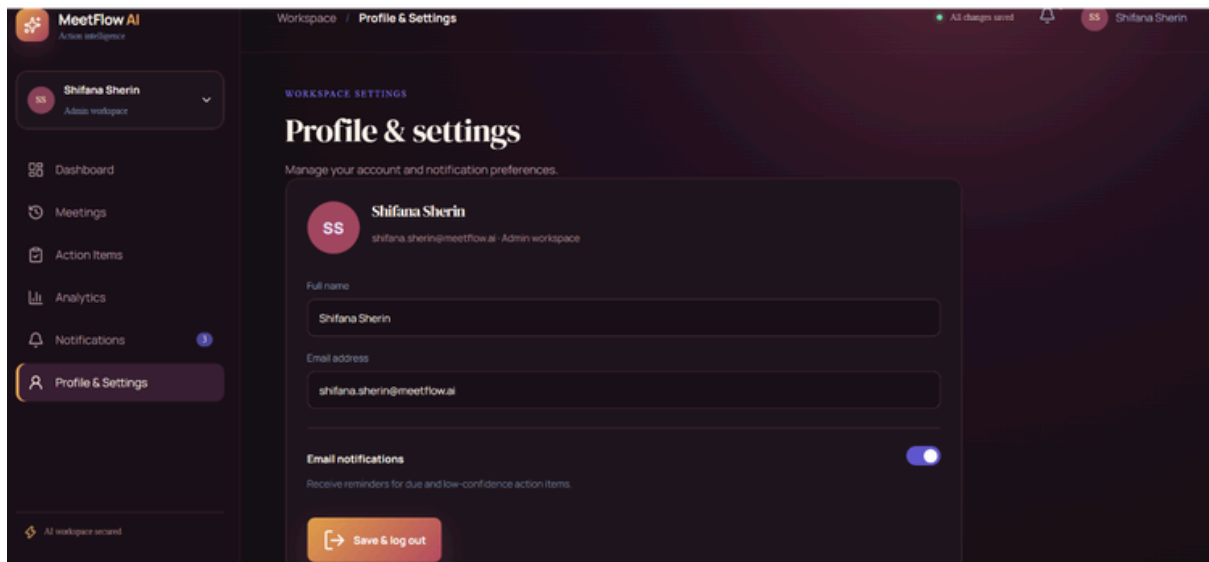
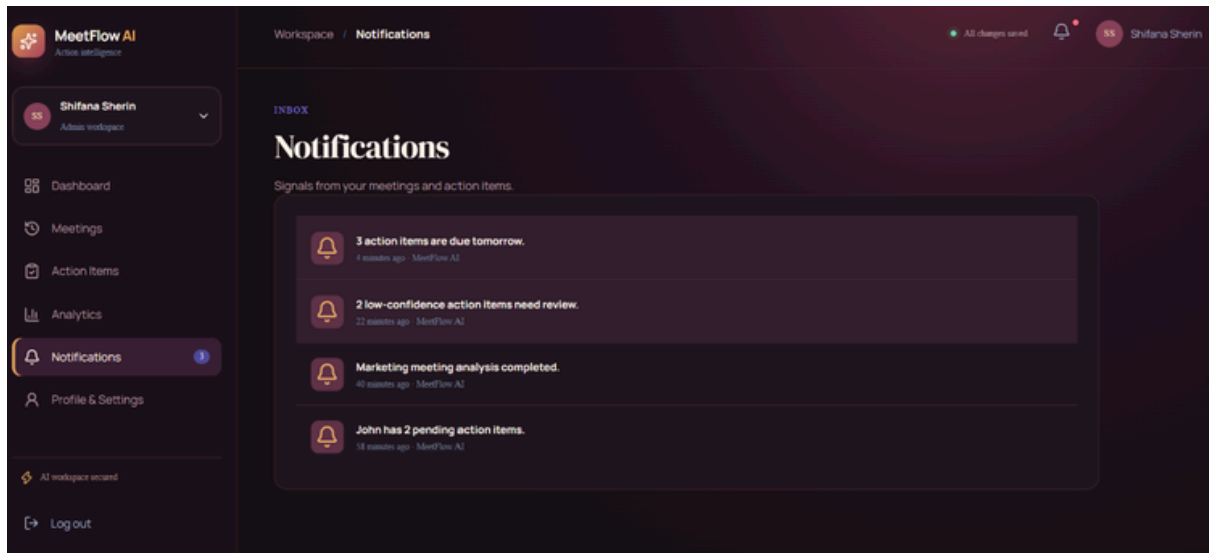
⌚ No deadline detected.

🗑️ Delete

Cancel

Save changes





13. Future Scope

- **Voice-Based Meeting Analysis** – The system can directly process recorded audio or video meetings without requiring a transcript.
- **Multilingual Support** – It can be extended to understand multiple languages and mixed-language conversations.
- **Advanced AI Models** – More powerful NLP and transformer models can improve task and context understanding.

- **Calendar Integration** – Extracted deadlines can be automatically added to Google Calendar or other scheduling platforms.
- **Email and Messaging Integration** – Follow-up reminders can be sent automatically through email or messaging applications.
- **Smart Priority Prediction** – AI can predict task priority based on deadlines, urgency, and meeting context.
- **Voice Reminder Generation** – The system can generate personalized voice reminders for pending tasks.
- **Advanced Team Analytics** – Future versions can provide detailed workload and productivity analytics.
- **Real-Time Extraction** – Action items can be identified while the meeting is still taking place.
- **Enterprise Integration** – The system can be integrated with project-management and collaboration platforms for complete task management.

14. Conclusion

The **AI Meeting Action-Item Extractor** provides an intelligent approach for converting unstructured meeting conversations into structured and actionable information. By using **Artificial Intelligence and Natural Language Processing**, the system can identify tasks, responsible persons, deadlines, status, and priorities from meeting transcripts.

The inclusion of **confidence scoring, validation, duplicate detection, cross-meeting tracking, overdue-task identification, and human verification** improves the reliability and usefulness of the extracted information. The interactive dashboard further helps users monitor workloads, task progress, and pending

responsibilities.

Overall, the system reduces manual effort, minimizes missed action items, and improves accountability after meetings. It transforms meeting transcripts from passive records into useful task-management information that can support better collaboration and project follow-up.

15 REFERENCE

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Bird, S., Klein, E. and Loper, E., **Natural Language Processing with Python**, O'Reilly Media.

Manning, C. D., Raghavan, P. and Schütze, H., **Introduction to Information Retrieval**, Cambridge University Press.

Vaswani, A. et al., **Attention Is All You Need**, Advances in Neural Information Processing Systems (NeurIPS), 2017.

[Python Documentation](#)

[spaCy Documentation](#)

[Scikit-learn Documentation](#)

[Streamlit Documentation](#)

